
JOINT DISTRIBUTION-INFORMED SHAPLEY VALUES FOR SPARSE COUNTERFACTUAL EXPLANATIONS

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ABSTRACT

Counterfactual explanations (CE) aim to reveal how small input changes flip a model’s prediction, yet across the many settings in which they are deployed a recurring failure mode is that they modify more features than necessary, reducing clarity and actionability. We introduce *COLA*, a broadly applicable, model- and generator-agnostic post-hoc framework that refines essentially any given CE by computing a coupling via optimal transport (OT) between factual and counterfactual sets and using it to drive a Shapley-based attribution (*p-SHAP*) that selects a minimal set of edits while preserving the target effect. Because the framework makes no structural assumptions about the predictor or the CE generator, it applies uniformly across model families and across instance-, group-, and distribution-level CE paradigms. Theoretically, OT minimizes an upper bound on the W_1 divergence between factual and counterfactual outcomes and, under mild conditions, refined counterfactuals are guaranteed not to move farther from the factuals than the originals. Empirically, across a diverse evaluation spanning four datasets from distinct application domains, twelve model classes, and five representative CE generators, COLA consistently attains the same target effects while using only 26–45% of the original feature edits, and on a tractable benchmark it approaches the exact optimum.

The experiment code and software will be released upon acceptance.

1 BACKGROUND

Explainable Artificial Intelligence (XAI) is essential for making artificial intelligence systems transparent and trustworthy across the full range of domains in which they are deployed (Arrieta et al., 2020; Das & Rad, 2020). Within this broad area, feature attributions (FA) methods, such as Shapley values (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017), quantify how much each input feature contributes to a machine learning (ML) model’s output, and thereby help simplify complex models of many kinds by surfacing the most influential features. In healthcare, for example, Shapley values can isolate key factors such as age and medical history and so assist clinicians in interpreting model decisions (Ter-Minassian et al., 2023; Nohara et al., 2022), but the same principle carries over to finance, criminal justice, and other high-stakes settings. A complementary technique, counterfactual explanations (CE) (Wachter et al., 2017; Guidotti, 2022), shows how small changes in input features can lead to different outcomes. Because hundreds of CE algorithms have now been proposed (Guidotti, 2022; Verma et al., 2020), and because each is tailored to its own scenarios, goals, and tasks, no single CE algorithm can be expected to serve every use case. The objective of one algorithm may be to find a single counterfactual instance for each factual instance; another may treat multiple instances as a group and seek one or several counterfactuals for each observation. In some settings the focus is the entire dataset, where the aim is to identify global CE that indicate a common direction to move factual instances toward the desired model output (Rawal & Lakkaraju, 2020; Ley et al., 2022; 2023; Carrizosa et al., 2024); in others the group of factual instances is viewed as a distribution, and the goal is a counterfactual distribution that preserves the shape of the factual distribution (You et al., 2025) while incurring comparable cost. Assumptions about the predictor vary just as widely: some CE algorithms require differentiable models, whereas others are designed for tree-based or ensemble models. This diversity of tasks, granularities, and model assumptions is precisely the setting our work is meant to span.

Problem Description and Challenges We address a broad and general problem in this paper, one that builds on the extensive foundations established across the literature (see Appendix A). To be specific, we seek to answer the following question: Although both FA and CE are vital for making AI models more interpretable and accountable, relying on either one alone has drawbacks that recur across application settings. FA alone may not translate into tangible steps that a stakeholder can act on, and CE alone may include unnecessary feature changes that are impractical to enact. Therefore, we address a broad and general problem in this paper, one that builds on the extensive foundations established across the literature (see Appendix A). To be specific, we seek to answer the following question:

Given a (group of) factual instance(s), how can we devise an action plan that requires the least feature modifications to achieve a desired counterfactual outcome?

$\mathbf{x}$	$\mathbf{z}'$	$\mathbf{z}''$	$\mathbf{y}^*$
$\textcircled{a}$ $\blacktriangleleft$ $\textcircled{R}$	$\textcircled{a}$ $\blacktriangleleft$ $\textcircled{R}$	$\textcircled{a}$ $\blacktriangleleft$ $\textcircled{R}$	$\textcircled{R}$
200 5 No	250 8 Yes	200 5 No	Yes
150 3 No	150 3 No	150 7 Yes	Yes
100 2 No	350 9 Yes	100 2 No	Yes
150 6 No	150 6 No	350 6 Yes	Yes

Figure 1: [A representative example: user engagement on an e-commerce platform] We use this scenario to illustrate a problem that arises identically in many other domains. A platform aims to increase user registrations and has collected data on user interactions, such as the amount of money spent ($\textcircled{a}$), the number of clicks ($\blacktriangleleft$), and whether the user has registered ($\textcircled{R}$). In the original data ($\mathbf{x}$), no users are registered. Action plans $\mathbf{z}'$ and $\mathbf{z}''$ adjust user characteristics to reach the desired outcome ($\mathbf{y}^*$) of full registration. Both plans achieve a half counterfactual effect, yet $\mathbf{z}''$ requires fewer modifications than $\mathbf{z}'$ — the kind of edit-minimal action plan that is preferable regardless of the underlying application.

Three major challenges must be met if a solution is to remain broadly applicable. First, because the problem is inherently task-specific, it is unrealistic to expect a single CE algorithm to meet all needs universally, so a general solution must instead operate on top of whatever CE generator a practitioner already uses. Second, the approach should avoid strong assumptions about the model — for instance, differentiability or other special structure — so that it applies across the widest possible range of model families. Third, FA scores such as feature importance can be misleading whenever they are not coherent with the changes needed for the counterfactual effect. In other words, it is generally ineffective to perform FA independently of CE to select the most important features to change. We demonstrate later that this decoupling can lead to counterproductive feature modifications (also referred to as actions from users/stakeholders) in Result II of Section 6, since the features deemed important in general need not align with the specific pathways that achieve a desired counterfactual outcome.

Main Contributions We introduce COUNTERFACTUALS WITH LIMITED ACTIONS (COLA), a general post-hoc framework that refines CE across arbitrary models and CE generators by using an optimal transport (OT)–induced coupling between factual and counterfactual sets to guide Shapley attributions. This yields p -SHAP, which unifies a range of commonly used Shapley methods as special cases under appropriate couplings and exposes a modular interface for attribution and edit selection that is agnostic to the choice of predictor and generator. We provide theoretical guarantees that hold under broad conditions: OT minimizes an upper bound on the W_1 divergence, and under mild assumptions refined counterfactuals are provably no farther from the factuals than the originals. Empirically, across an evaluation that spans four datasets from distinct domains, twelve model classes, and five CE methods covering instance-, group-, and distribution-level paradigms, COLA with p -SHAP consistently attains the same target effects while using only 26–45% of the original feature edits, and on a tractable benchmark it comes close to the exact optimum.

To the best of our knowledge, this is the first method to systematically address the problem in Figure 1 without committing to any particular CE algorithm or model class, and it is therefore positioned to serve as a general-purpose refinement layer wherever counterfactual explanations are used.

2 PROBLEM FORMULATION

We now give a formal statement of the problem described in Figure 1, kept deliberately general so that it subsumes the many instantiations discussed above. Denote by $f : \mathbb{R}^d \rightarrow \mathbb{R}$ an arbitrary black-box ML model. Denote by $\mathbf{x}$ any observed (factual) data with n rows and d columns ($\mathbf{x} \in \mathbb{R}^{n \times d}$, $n \geq 1$, and $d \geq 1$), so that a single instance and a whole group of instances are handled by the same notation. Let $\mathbf{y}^*$ be the target model output ($\mathbf{y}^* \in \mathbb{R}^m$, $m \geq 1$). The optimization seeks a (group of) counterfactual data instance(s) $\mathbf{z}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$), subject to a maximum number of allowed feature changes C , so as to drive the model output(s) as close as possible to $\mathbf{y}^*$. Let D denote a divergence function that measures the dissimilarity between two entities. The problem is stated below.

$$\min_{\mathbf{c}, \mathbf{z}} D(f(\mathbf{z}), \mathbf{y}^*), \quad (1a)$$

$$\text{s.t. } D(\mathbf{z}, \mathbf{x}) \leq \epsilon, \quad (1b)$$

$$\sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (1c)$$

$$z_{ik} \leq x_{ik}(1 - c_{ik}) + M c_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d, \quad (1d)$$

$$z_{ik} \geq x_{ik}(1 - c_{ik}) - M c_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (1e)$$

The objective equation (1a) and the constraint equation (1b) recover the typical CE optimization in its general form: $\mathbf{z}$ is expected to make $f(\mathbf{z})$ close to $\mathbf{y}^*$ while staying close to $\mathbf{x}$, so that $\mathbf{z}$ can serve as a contrastive reference explaining why $f(\mathbf{x})$ does not reach $\mathbf{y}^*$. We deliberately place no restriction on the function D , keeping it as general as possible; admissible choices for D include Euclidean distance, OT, maximum mean discrepancy (MMD), and differences of the model outcome in mean or median, and other divergences fit the same template. The remaining constraints equation (1c)–equation (1e) then augment this general CE optimization with an explicit budget on actions. On top of the counterfactual data $\mathbf{z}$, we also optimize an indicator variable $\mathbf{c}$, such that z_{ik} is not allowed to change iff $c_{ik} = 0$. Maximum C changes are allowed as imposed by equation (1c). Inspecting equation (1d) and equation (1e), if $c_{ik} = 0$, x_{ik} equals z_{ik} and no changes happen at (i, k) . Otherwise, if $c_{ik} = 1$, remark that M is a sufficiently large constant such that z_{ik} has good freedom to change.

To solve equation (1), we resort to FA to identify the most influential features and thereby obtain the modification indicator variable $\mathbf{c}$. The next section reviews commonly used Shapley value methods for FA, which, together with the one we later propose, are integrated into our algorithmic framework COLA to obtain the refined counterfactual $\mathbf{z}$. The problem is computationally difficult in general — indeed it remains hard even in the restricted case $d = 1$ with linear models, see Appendix B — which further motivates a broadly applicable approximate framework rather than a special-purpose exact solver.

3 PRELIMINARIES ON SHAPLEY VALUE

This section reviews commonly used Shapley value methods for FA, which, together with the one we later propose, are integrated into our algorithmic framework COLA. We first recall the concept of Shapley value from game theory, and then discuss the range of Shapley methods that are widely used for FA. A comprehensive overview of the different Shapley value methods can be found in (Sundararajan & Najmi, 2020).

In cooperative game theory, a coalitional game is characterized by a finite set of players (in our context, features), denoted by $\mathcal{F} = \{1, 2, \dots, d\}$, and a characteristic function v . This function $v : 2^{\mathcal{F}} \rightarrow \mathbb{R}$ maps each subset $S \subseteq \mathcal{F}$ to a real number $v(S)$, representing the total payoff that can be achieved by the members of S through cooperation, with the condition $v(\emptyset) = 0$. The Shapley value, as defined based on v later, is a fundamental concept in this framework, providing an equitable method to distribute the overall payoff, $v(\mathcal{F})$, of the grand coalition among the individual players.

Prior to defining the Shapley value, denote $\Delta(k, S)$ the incremental value produced by a player (feature) k to a coalition S , aligning with the concept of marginal contribution. This is defined as the additional payoff realized by the coalition due to the inclusion of the player k , i.e.,

$$\Delta(k, S) = v(S \cup k) - v(S). \quad (2)$$

The Shapley value ϕ_i for a player i in a cooperative game is defined in equation (3) below.

$$\phi_k = \frac{1}{d} \sum_{S \subset \mathcal{F} \setminus \{k\}} \binom{d-1}{|S|}^{-1} \Delta(k, S). \quad (3)$$

The value ϕ_k is obtained by averaging the normalized marginal contributions $\Delta(k, S)$ of feature k across all subsets S of the feature set that exclude k . The normalization factor is the inverse of the binomial coefficient $\binom{d-1}{|S|}$, where d is the total number of features. Summing over all subsets and dividing by d yields a fair and axiomatic distribution of payoffs among players, reflecting each feature’s contribution relative to the whole collection of features — a property that holds independently of the particular model or task at hand.

For an arbitrary data point $\mathbf{x}_i$, we use $\mathbf{x}_{i,S}$ to represent its part containing only the features in S , and $\mathbf{x}_{i,\mathcal{F} \setminus S}$ the other remaining part. The same rule applies for other notations as well. Below, we introduce commonly used Shapley value methods.

Baseline Shapley (B-SHAP). This approach relies on a specific baseline data point $\mathbf{r}_j$ as the reference for any $\mathbf{x}_i$. The set function is defined to be

$$v_B^{(i)}(S) = f(\mathbf{x}_{i,S}; \mathbf{r}_{j,\mathcal{F} \setminus S}) - f(\mathbf{r}_j), \quad (4)$$

where a feature’s absence is modeled using its value in the reference baseline data point $\mathbf{r}$ (Lundberg & Lee, 2017; Sun & Sundararajan, 2011; Merrick & Taly, 2020). This method rests on the comparatively strong assumption of an exact alignment between $\mathbf{x}$ and $\mathbf{r}$.

Random Baseline Shapley (RB-SHAP). This approach is a variant of BShap and is implicitly used in (Lundberg & Lee, 2017) as well as the famous “SHAP” library. The equation is as follows.

$$v_{RB}^{(i)}(S) = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}_{i,S}; \mathbf{x}'_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}')]. \quad (5)$$

where $\mathcal{D}$ is the background distribution that is commonly selected to be the training dataset (Lundberg & Lee, 2017; Merrick & Taly, 2020).

Counterfactual Shapley (CF-SHAP). This is an extension from RB-SHAP, by taking the distribution $\mathcal{D}$ as the counterfactual distribution defined on every single factual instance $\mathbf{x}_i$ ($i = 1, 2, \dots, n$).

$$v_{CF}^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{r})]. \quad (6)$$

This method has been adopted in (Albini et al., 2022; Kommiya Mothilal et al., 2021) and shown to have advantages for contrastive FA. It relaxes the exact-alignment requirement to a probabilistic alignment between $\mathbf{x}$ and $\mathbf{r}$, though, as we discuss later, that alignment must still be supplied by the CE generation mechanism.

4 THE PROPOSED JOINT-PROBABILITY-INFORMED SHAPLEY (p -SHAP) AND ITS THEORETICAL ASPECTS

(Proposed) p -SHAP We generalize equation (4) to equation (6) by integrating an algorithm A_{Prob} that returns their joint probability.¹ Our p -SHAP is defined as follows,

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r}|\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r})} [f(\mathbf{r})], \quad (7a)$$

$$\text{s.t. } p = A_{\text{Prob}}(\mathbf{x}, \mathbf{r}). \quad (7b)$$

Crucially, and in contrast to prior methods that rely on random baselines or on a generator-supplied alignment, p -SHAP resolves the alignment problem intrinsically by identifying an optimal coupling via OT. This reframes feature attribution not merely as a prediction decomposition but, more generally, as a transport problem that minimizes the cost of explanation — a viewpoint that applies to any pair of factual and counterfactual sets regardless of how the latter was produced.

¹First, p -SHAP degrades to B-SHAP in equation (4) when A_{Prob} defines a joint distribution between $\mathbf{x}$ and $\mathbf{r}$ that indicates an $i \leftrightarrow j$ alignment of for any $\mathbf{x}_i, \mathbf{r}_j$. Second, p -SHAP degrades to RB-SHAP in equation (5) when A_{Prob} is defined to be independent of CE but associates with an arbitrary distribution $\mathcal{D}$. Third, p -SHAP degrades to CF-SHAP in equation (6) when A_{Prob} is built upon a known distribution of CE.

Remark that A_{Prob} is independent of the CE algorithm and depends only on the generated counterfactual $\mathbf{r}$. By depending solely on these fixed components, p -SHAP delivers consistent FA that is insulated from the variability of different CE generation processes — a decoupling that lets a single attribution recipe carry across the whole spectrum of CE generators, and a major point of departure from CF-SHAP. Contrary to common expectations, we demonstrate that OT can be more effective than relying on a counterfactual distribution defined by a specific CE generation mechanism, as done by Albini et al. (2022), in Result II of Section 6 later.

A central instance considered in this paper is the OT problem (equivalently, the 2-Wasserstein divergence) defined below, and the transportation plan $\mathbf{p}_{\text{OT}}$ obtained by solving it is used as the joint distribution of $\mathbf{x}$ and $\mathbf{r}$ in p -SHAP. We stress, however, that A_{Prob} is a general slot in the framework: OT is a convenient and effective default, not a restriction, and any coupling that reflects the factual–counterfactual relationship can be substituted.

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 + \varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log \left(\frac{p_{ij}}{\mu_i \nu_j} \right). \quad (8)$$

Here $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$ denote the marginal distributions of $\mathbf{x}$ and $\mathbf{r}$ respectively, and $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ the set of joint distributions (i.e., all admissible transport plans). The term $\varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log(p_{ij}/(\mu_i \nu_j))$ is the entropic regularization with coefficient $\varepsilon \geq 0$; taking $\varepsilon > 0$ accelerates the computation of OT and makes the framework practical at the scales encountered across our datasets.

Theoretical Aspects of p -SHAP OT minimizes the total feature modification cost (i.e., the cost of moving $\mathbf{x}$ toward $\mathbf{r}$) under the alignment it induces between factual $\mathbf{x}$ and counterfactual $\mathbf{r}$. This corresponds directly to our objective of finding feature modifications that achieve the counterfactual outcomes at minimal cost. This connection can be made rigorous under only a Lipschitz-continuity assumption on the predictive model f — an assumption satisfied, at least locally, by a wide range of models rather than any single family. In Theorem 4.1 below (proof in Appendix C), we establish that the transportation plan $\mathbf{p}_{\text{OT}}$ used in p -SHAP minimizes an upper bound on the divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$. Specifically, the 1-Wasserstein distance between $f(\mathbf{x})$ and $\mathbf{y}^*$, is bounded by the Lipschitz constant (assuming Lipschitz continuity of f) multiplied by the square root of the minimized expected cost of changing $\mathbf{x}$ towards $\mathbf{r}$, i.e. $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$ where p_{ij} ($j = 1, 2, \dots, m$) quantify how the feature values of $\mathbf{x}_i$ should be adjusted towards those of one or multiple $\mathbf{r}_j$. Practically, this means that in p -SHAP, the OT plan $\mathbf{p}_{\text{OT}}$ provides a strategy to adjust the feature values of $\mathbf{x}$ towards those of $\mathbf{r}$ in a way that minimizes the expected modification cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$. Compared to other modification plans ($\mathbf{p} \in \Pi$), $\mathbf{p}_{\text{OT}}$ yields the minimal possible cost, which in turn provides the tightest upper bound on the violation of the counterfactual effect $W_1(f(\mathbf{x}), \mathbf{y}^*)$ in proportion to this cost.

Theorem 4.1 (*p -SHAP Towards Counterfactual Effect*). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f: \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{\text{OT}}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi.$$

Namely, $\mathbf{p}_{\text{OT}}$ minimizes the upper bound of $W_1(f(\mathbf{x}), \mathbf{y}^*)$, where the upper bound is based on the expected feature modification cost.

Although Theorem 4.1 bounds the transport cost, this bound serves as a convex proxy for the $\mathcal{NP}$ -hard discrete sparsity (L_0) problem that arises in the general setting. By concentrating attribution mass on the most efficient paths, the OT coupling reliably guides the greedy selection in Algorithm 1 to discard non-essential features, and does so without appeal to any model- or generator-specific structure. In addition, p -SHAP is conceptually correct in attributing the causal behavior to the modifications of the characteristics, stated in Theorem 4.2 below (proof in Appendix D).

Theorem 4.2 (*Interventional Effect of p -SHAP*). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the*

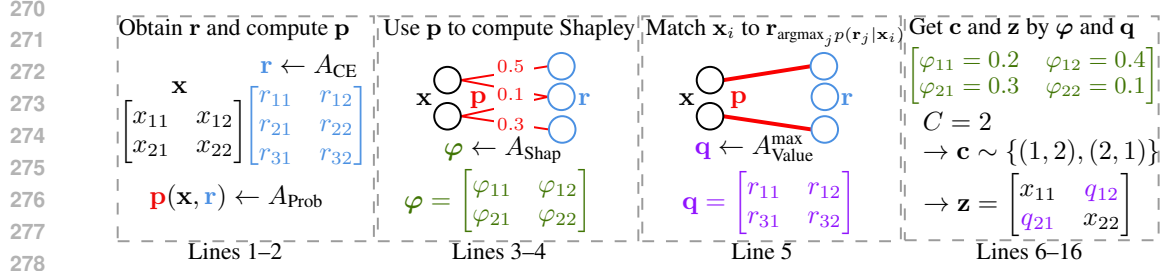


Figure 2: [An illustration of COLA] This figure shows how COLA obtains $\mathbf{c}$ and $\mathbf{z}$ for equation (1). For simplicity of illustration we use $A_{\text{Value}}^{\max}$ in line 5; the other value-assignment choices proceed analogously. In lines 6–16, we assume $C = 2$, and the sampling yields exactly two positions for modification according to the probability matrix φ .

intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Furthermore, we note that p -SHAP preserves the desirable axioms of B-SHAP and RB-SHAP, which makes it a sound and broadly usable attribution tool. We omit the proof and refer to (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017) for the axioms of Shapley.

5 THE ALGORITHMIC FRAMEWORK COLA

Sketch The algorithmic framework COLA, stated in Algorithm 1 below, targets equation (1) and is built from four interchangeable categories of algorithms. Because each category is a pluggable module rather than a fixed choice, the framework accommodates a wide variety of CE generators, couplings, attribution schemes, and value-assignment strategies within one unified procedure. We explain Algorithm 1 in details below, along with an illustration in Figure 2.

Algorithm 1 COUNTERFACTUALS WITH LIMITED ACTIONS (COLA)

Require: Model f , factual $\mathbf{x} \in \mathbb{R}^{n \times d}$, target $\mathbf{y}^* \in \mathbb{R}^m$, ϵ , and C
Ensure: Action plan $\mathbf{c} \in \mathbb{R}^{n \times d}$ and correspondingly a refined counterfactual $\mathbf{z} \in \mathbb{R}^{n \times d}$

- 1: Use $A_{\text{CE}}(f, \mathbf{x}, \mathbf{y}^*, \epsilon)$ to obtain $\mathbf{r} \in \mathbb{R}^{m \times d}$, with $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$.
- 2: Use $A_{\text{Prob}}(\mathbf{x}, \mathbf{r})$ to obtain the joint distribution matrix $\mathbf{p} \in \mathbb{R}_+^{n \times m}$.
- 3: Use $A_{\text{Shap}}(\mathbf{x}, \mathbf{r}, \mathbf{p})$ to obtain the shapley values $\phi \in \mathbb{R}^{n \times d}$ for each element of $\mathbf{x}$.
- 4: Normalize the element-wise absolute values of ϕ , i.e., $\varphi_{ik} \leftarrow |\phi_{ik}| / \|\phi\|_1$ ($\varphi \in \mathbb{R}_+^{n \times d}$).
- 5: Use $A_{\text{Value}}(\mathbf{r}, \mathbf{p})$ to obtain matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$.
- 6: For $\mathbf{c} \in \{0, 1\}^{n \times d}$, $c_{ik} \leftarrow 0$ ($i = 1, \dots, n$, $k = 1, \dots, d$).
- 7: Sample C pairs (i, k) according to the probability matrix φ , and let $c_{ik} = 1$ for them.
- 8: Let $\mathbf{z} \leftarrow \mathbf{x}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$).
- 9: **for** $i \leftarrow 1$ to n **do**
- 10: **for** $k \leftarrow 1$ to d **do**
- 11: **if** $c_{ik} = 1$ **then**
- 12: $z_{ik} \leftarrow q_{ik}$
- 13: **end if**
- 14: **end for**
- 15: **end for**
- 16: **return** $\mathbf{c}$ and $\mathbf{z}$

Line 1 (Applying a CE algorithm to find a counterfactual $\mathbf{r}$). The CE algorithm A_{CE} takes the model f , the factual $\mathbf{x}$, the target outcome $\mathbf{y}^*$, and the tolerance ϵ as input, and returns a counterfactual $\mathbf{r}$ that stays close to $\mathbf{x}$, with $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$. Any CE generator can occupy this slot, whether it produces per-instance, group, or distributional counterfactuals.

Line 2 (*Seeking a joint distribution of $\mathbf{x}$ and $\mathbf{r}$*). We use an algorithm A_{Prob} for this task, which takes $\mathbf{x}$ and $\mathbf{r}$ as input and outputs a matrix representing the joint distribution over all n and m data points in $\mathbf{x}$ and $\mathbf{r}$, respectively. This joint distribution $\mathbf{p}$ encodes an alignment (or matching) between the factual and counterfactual rows, and we use it in Line 5 to compute the values from which $\mathbf{z}$ is later composed. As discussed in Section 3, when the row alignment is unknown — the typical situation for group and distributional CE — A_{Prob} can be based on OT to compute a joint distribution attaining the smallest OT distance between $\mathbf{x}$ and $\mathbf{r}$; when a trustworthy alignment is available, any joint distribution that faithfully reflects the alignment between the rows of $\mathbf{x}$ and $\mathbf{r}$ may be used instead. This flexibility is what lets a single instantiation of A_{Prob} cover both the aligned and unaligned regimes.

Lines 3–4 (*p-SHAP FA*). We apply equation (7) to compute the Shapley values for $\mathbf{x}$. The joint distribution $\mathbf{p}$ may be used here (though it is not required) to align each row of $\mathbf{x}$ with its corresponding counterfactual rows of $\mathbf{r}$, so that the selected rows of $\mathbf{r}$ act as the most representative contrastive reference for that row of $\mathbf{x}$. Numerically, this alignment has a substantial effect on the contrastive FA, and the effect is consistent across the models and generators we consider. The Shapley values of $\mathbf{x}$ (as a matrix) are then taken element-wise in absolute value and normalized to sum to one. The resulting matrix φ constitutes our FA.

Line 5 (*Computing feature values*). The algorithm A_{Value} handles this task, taking the counterfactual $\mathbf{r}$ and the joint distribution $\mathbf{p}$ as input. For any row i of $\mathbf{x}$, A_{Value} selects one or several rows of $\mathbf{r}$ to serve as references for changing $\mathbf{x}$, and returns a matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$ in which each element q_{ik} is a counterfactual candidate for x_{ik} ($\forall i, k$). We describe two representative instances below, $A_{\text{Value}}^{\text{max}}$ and $A_{\text{Value}}^{\text{avg}}$, covering respectively the single-row and multiple-row cases; other value-assignment rules fit the same interface.

For any row $\mathbf{x}_i$, $A_{\text{Value}}^{\text{max}}$ selects the row of $\mathbf{r}$ with the highest probability.

$$\mathbf{q} = A_{\text{Value}}^{\text{max}}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = r_{\tau(i),k} \text{ and } \tau(i) = \arg \max_{j=1,2,\dots,m} p_{ij}. \quad (9)$$

The algorithm $A_{\text{Value}}^{\text{avg}}$ computes q_{ik} as a convex combination as a weighted average of $r_{1k}, r_{2k}, \dots, r_{mk}$.

$$\mathbf{q} = A_{\text{Value}}^{\text{avg}}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}. \quad (10)$$

Lines 6–16. Recall that the non-negative matrix φ is normalized to sum to one. We can therefore treat it as a policy for selecting the positions in $\mathbf{x}$ to be replaced (i.e., $c_{ik} = 1$), which is what line 7 does. For every i, k with $c_{ik} = 1$, x_{ik} is modified to q_{ik} ; the modified matrix is returned as $\mathbf{z}$, and together with $\mathbf{c}$ it forms the optimized solution to equation (1). Nothing in these steps depends on the specific predictor or CE generator, so the same selection procedure applies uniformly across settings.

By Theorem 4.1, COLA is designed to minimize the dissimilarity between $f(\mathbf{z})$ and $\mathbf{y}^*$ by modifying $\mathbf{z}$ based on feature attribution results, which identify the most important features to adjust to achieve the desired counterfactual effect. The theorem below (proof in Appendix E) shows that the refined $\mathbf{z}$ produced by the COLA framework satisfies the constraint equation (1b), with the Frobenius norm as the distance measure, and it does so for arbitrary dimensions rather than only the special case $n = m$. Empirical evidence supporting the broad applicability of this conclusion can be found in Table 3.

Theorem 5.1 (Counterfactual Proximity). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{Value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is strictly closer (or equal) to the factual input $\mathbf{x}$ compared to the aligned counterfactual proposal $\mathbf{q}$, ensuring that the refinement process introduces no additional divergence.

Corollary 5.2. *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

recovering the intuition that the refined counterfactual is at least as close to the facts as the original counterfactual set reordered by σ .

Complexity of COLA Let $\mathcal{O}(M_{\text{CE}})$ denote the complexity of A_{CE} , which may be any CE generator. For A_{Shap} , consider estimating Shapley values by weighted linear regression, and let M_{Shap} be the number of sampled subsets. The overall complexity of COLA with respect to n , m , d , and the entropic-OT regularization parameter ε is $\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N$ where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ if $A_{\text{Value}}^{\max}$ is used and $N = \mathcal{O}(nmd)$ if $A_{\text{Value}}^{\text{avg}}$ is used. See Appendix F for more details.

6 NUMERICAL RESULTS

This section evaluates the effectiveness of COLA on the problem in equation (1), with $\mathbf{y}^* = f(\mathbf{r})$ where $\mathbf{r}$ is the counterfactual obtained from a CE method A_{CE} . To probe robustness across different notions of divergence, we adopt four functions spanning distinct statistical granularities: OT measures the distance between entire distributions; MMD measures the divergence between distribution means in a high-dimensional feature space; and absolute mean difference (MeanD) and absolute median difference (MedianD) measure the divergence in mean and median, respectively. The numerical study is designed to show: I) *COLA’s effectiveness for modification minimality*. II) *p-SHAP’s superior performance than other Shapley methods towards modification minimality*. III) *COLA’s near-optimal performance*.

Experiment Setup To exercise the framework across a broad range of conditions, the experiments span 4 datasets drawn from distinct application domains (credit, hospitality, and criminal justice) for binary classification, 5 CE algorithms designed for diverse goals, and 12 classifiers, as summarized in Table 1; a combination of dataset, A_{CE} algorithm, and model defines an “experiment scenario”. Together these choices yield a large and heterogeneous collection of scenarios, so that the reported behavior reflects general performance rather than any single configuration.

Table 1: [Experiment Scenarios Setup] Four datasets from distinct domains are used to benchmark five CE algorithms over 12 model classes, giving broad coverage of tasks, generators, and predictors. A “scenario” is a combination of dataset, A_{CE} algorithm, and model f .

Dataset	HELOC (FICO, 2018), German Credit (Hofmann, 1994), Hotel Bookings (Antonio et al., 2019), COMPAS (Jeff Larson et al., 2016)
A_{CE}	DiCE (Mothilal et al., 2020), AReS (Rawal & Lakkaraju, 2020), GlobeCE (Ley et al., 2023), KNN (Albini et al., 2022; Contardo et al.; Forel et al., 2023), Discount (You et al., 2025)
Model f	Bagging, LightGBM, Support Vector Machine (SVM), Gaussian Process (GP), Radial Basis Function Network (RBF), XGBoost, Deep Neural Network (DNN), Random Forest (RndForest), AdaBoost, Gradient Boosting (GradBoost), Logistic Regression (LR), Quadratic Discriminant Analysis (QDA)

Table 2: [Experiment Methods Setup] The table defines the 6 methods compared, colored to match Figure 3. Each method is evaluated across the experiment scenarios of Table 1, so that comparisons reflect behavior over the full range of settings rather than a single one.

Method	The probability $\mathbf{p}$ used by A_{Value} and A_{Shap}	The Shapley algorithm A_{Shap}
RB- p_{Uni}	$\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm \ (\forall i, j)$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
RB- p_{OT}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8) (but not used in } A_{\text{Shap}})$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
CF- p_{Uni}	$\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm \ (\forall i, j)$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{Rnd}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched randomly to an } \mathbf{r}_j$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{OT}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8)}$	p -SHAP with $\mathbf{p}$
CF- p_{Ect}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched to known counterpart } \mathbf{r}_j$	(CF or B)-SHAP, $\mathcal{D}(\mathbf{x}_i) \rightarrow \mathbf{r}_j$

We briefly introduce the A_{CE} methods in Table 1, chosen to represent the main CE paradigms. DiCE and KNN are instance-based CE methods, producing counterfactual(s) for each factual instance individually. AReS and GlobeCE are group-based CE methods, finding a collection of counterfactuals for the whole factual set treated as a group. Discount treats the factual instances as an empirical

distribution and seeks a counterfactual distribution that stays close to it. Spanning the instance-, group-, and distribution-level regimes in this way lets us assess COLA under the full breadth of CE formulations rather than a single one.

Table 2 defines the 6 methods, of which CF- p_{OT} is the proposed p -SHAP and the rest are baselines. Each is evaluated comprehensively across the many experiment scenarios of Table 1. A method is determined by a combination of the three algorithms A_{Prob} , A_{Value} , and A_{Shap} . For example, the first row, RB- p_{Uni} , uses a uniform distribution as A_{Prob} to compute $\mathbf{p}$, feeds the resulting $\mathbf{p}$ into equation (9) or equation (10) of A_{Value} to compute $\mathbf{q}$, and takes A_{Shap} to be RB-SHAP.

These methods are deliberately constructed to support controlled ablations that isolate the effect of each design choice. First, RB- p_{Uni} and CF- p_{Uni} differ only in that the latter uses CE information while the former does not. Second, CF- p_{Uni} , CF- p_{Rnd} , and CF- p_{OT} all use CE in FA but with different joint distributions, and we show later that the OT coupling outperforms the alternatives by a wide margin. Third, to confirm that OT helps because it informs A_{Shap} about the factual-counterfactual alignment rather than through some incidental factor, we compare CF- p_{OT} , which uses this alignment, with RB- p_{OT} , which does not. Finally, CF- p_{Ect} captures the special case in which each counterfactual has a known source that serves as an exact factual-counterfactual alignment, so that CF-SHAP coincides with B-SHAP. This spectrum of methods lets the conclusions generalize beyond any one alignment assumption.

Table 3: [COLA for Modification Minimality] The table reports the number of modified features in $\mathbf{z}$ for each method when $f(\mathbf{z})$ reaches an 80% counterfactual effect relative to $f(\mathbf{r})$. Each entry is averaged over 4 randomly selected scenarios from Table 1, with 100 runs per scenario, so that the numbers reflect typical behavior across settings rather than a hand-picked case. The symbol “-” indicates that the target counterfactual effect could not be achieved.

Dataset	Method	80% Counterfactual Effect		100% Counterfactual Effect	
		# Modified Features	$\frac{\ \mathbf{z}-\mathbf{x}\ }{\ \mathbf{r}-\mathbf{x}\ }$	# Modified Features	$\frac{\ \mathbf{z}-\mathbf{x}\ }{\ \mathbf{r}-\mathbf{x}\ }$
German Credit $ \mathcal{F} = 9$	RB- p_{Uni}	—	—	—	—
	RB- p_{OT}	5.29(± 0.09)	75.9%	—	—
	CF- p_{Uni}	—	—	—	—
	CF- p_{Rnd}	—	—	—	—
	CF- p_{OT}	1.70(± 0.02)	24.3%	3.13(± 0.03)	44.9%
Hotel Bookings $ \mathcal{F} = 29$	RB- p_{Uni}	7.10(± 0.08)	24.5%	—	—
	RB- p_{OT}	8.55(± 0.08)	50.2%	—	—
	CF- p_{Uni}	7.01(± 0.07)	41.1%	—	—
	CF- p_{Rnd}	10.63(± 0.08)	62.4%	—	—
	CF- p_{OT}	2.50(± 0.03)	14.6%	4.44(± 0.02)	26.0%
COMPAS $ \mathcal{F} = 15$	RB- p_{Uni}	5.02(± 0.05)	82.7%	—	—
	RB- p_{OT}	—	—	—	—
	CF- p_{Uni}	2.80(± 0.04)	34.4%	—	—
	CF- p_{Rnd}	2.58(± 0.04)	32.1%	—	—
	CF- p_{OT}	1.25(± 0.03)	14.8%	2.45(± 0.03)	30.0%
HELOC $ \mathcal{F} = 23$	RB- p_{Uni}	—	—	—	—
	RB- p_{OT}	—	—	—	—
	CF- p_{Uni}	—	—	—	—
	CF- p_{Rnd}	2.73(± 0.04)	15.7%	—	—
	CF- p_{OT}	2.35(± 0.03)	13.4%	7.745(± 0.05)	44.7%

Result I: COLA achieves substantial action reduction with only a minor loss in counterfactual effect In Table 3, we set a goal for $\mathbf{z}$ such that $f(\mathbf{z})$ reaches an 80% or 100% counterfactual effect relative to $f(\mathbf{r})$ ². Across these settings COLA consistently meets the goal while requiring far fewer actions in $\mathbf{z}$ (i.e., modifications of features in $\mathbf{z}$) than the original CE $\mathbf{r}$. In general, COLA attains the 80% counterfactual effect using only 13%–25% of the feature changes (measured by

²By definition $D(f(\mathbf{r}), \mathbf{y}^*) = 0$, which corresponds to a 100% counterfactual effect since $\mathbf{y}^* = f(\mathbf{r})$. An 80% counterfactual effect refers to the proportion of divergence removed by the refined CE $\mathbf{z}$, i.e., $Counterfactual\ Effect = 1 - D(f(\mathbf{z}), \mathbf{y}^*) / D(f(\mathbf{x}), \mathbf{y}^*) = 80\%$, with D being OT.

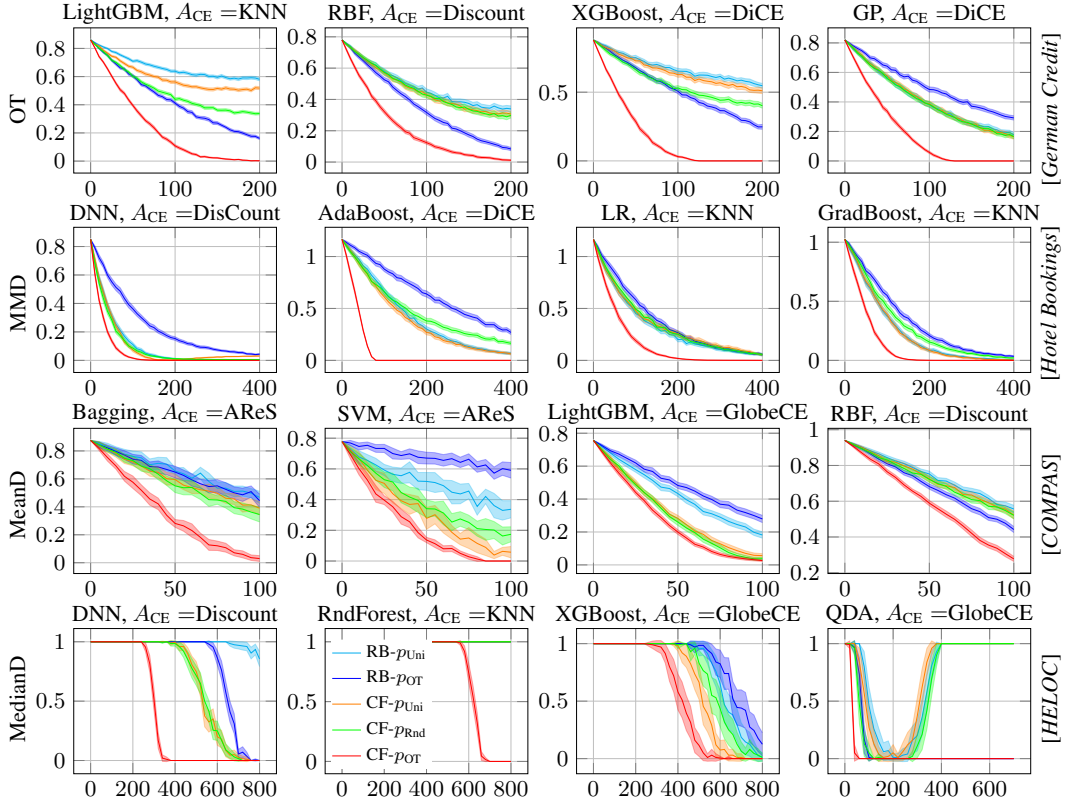


Figure 3: $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. allowed actions C , shown across datasets, models, and generators to illustrate the consistency of the trend. Experiments use 100 runs, with shaded regions giving the 99.9% confidence intervals. $A_{\text{Value}}^{\text{avg}}$ is used for *HELOC* and *COMPAS*, and $A_{\text{Value}}^{\text{max}}$ for *German Credit* and *Hotel Bookings*. The legend inside “RNdForest, $A_{\text{CE}} = \text{KNN}$ ” applies to all plots.

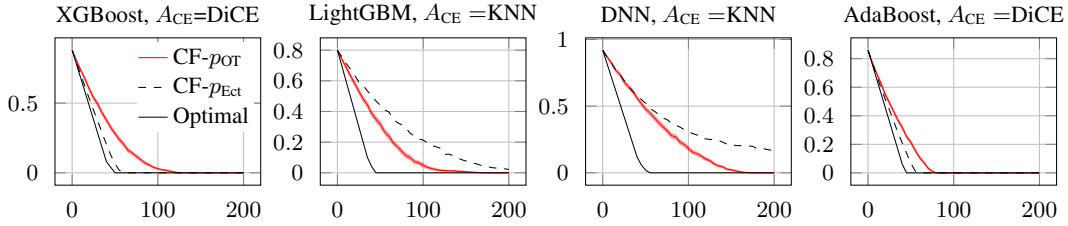


Figure 4: [*German Credit*] $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. allowed actions C , with D being *MeanD*, comparing COLA against the exact optimum on this tractable benchmark.

$\|\mathbf{z} - \mathbf{x}\| / \|\mathbf{r} - \mathbf{x}\|$). Moreover, COLA with p -SHAP (that is, $\text{CF-}p_{\text{OT}}$) is the only method able to reach the full 100% counterfactual effect, and it does so with just 26%–45% of the feature changes in the original $\mathbf{r}$. p -SHAP thus delivers the strongest action minimality, which we analyze in detail below.

Result II: p -SHAP outperforms the other Shapley methods in achieving the counterfactual effect Figure 3 evaluates the Shapley methods in equation (4)–equation (7), with the number of allowed feature changes C on the x-axis and the term $D(f(\mathbf{z}), \mathbf{y}^*)$ from equation (1) on the y-axis. Three observations emerge consistently across the panels. First, $\text{RB-}p_{\text{Uni}}$ and $\text{RB-}p_{\text{OT}}$ perform markedly worse than the rest, underscoring the general importance of using CE information when attributing features for modification minimality. Second, $\text{CF-}p_{\text{OT}}$ outperforms $\text{RB-}p_{\text{OT}}$, and since the two differ only in A_{Shap} (see Table 2), this isolates the benefit of OT to the effective factual–counterfactual alignment it supplies rather than to any incidental factor. Third, p -SHAP substantially outperforms $\text{CF-}p_{\text{Uni}}$ and $\text{CF-}p_{\text{Rnd}}$, confirming the value of the OT joint distribution: simply injecting

counterfactual information into FA (as $\text{CF-}p_{\text{Uni}}$ and $\text{CF-}p_{\text{Rnd}}$ do) is not enough, and a proper alignment between factual and counterfactual must be used — an alignment that, as Figure 4 later reveals, need not coincide with the one implied by the exact counterfactual generation mechanism.

COLA with p -SHAP performs well across a broad swath of the experiment scenarios in Table 1, as evidenced by both Figure 3 and Table 3, and an extensive body of additional results in Appendix H points the same way. Taken together, p -SHAP outperforms all the other Shapley methods for sparsity in CE across the settings we examine.

Result III: COLA is competitive with the mixed integer linear programming (MILP) optimum on tractable benchmarks This result shows how effectively p -SHAP removes any dependence on the CE generation process, by replacing the CE algorithm-dependent knowledge³ of $\mathcal{D}$ with the OT joint distribution between the factual and counterfactual data, as shown in Figure 4. We benchmark $\text{CF-}p_{\text{Ect}}$ within COLA, focusing on equation (1) with MeanD as the divergence D . Because $\text{CF-}p_{\text{Ect}}$ relies on a known factual-counterfactual alignment, this lets us assess how well COLA exploits such an alignment when one happens to be available. The theoretical optimum in this case is obtained by solving an MILP; see Appendix G for the derivation. Since the MILP is computationally heavy, we solve it only for *German Credit*, which suffices as a tractable reference point.

We find that $\text{CF-}p_{\text{Ect}}$ attains near-optimal performance using DiCE. For DiCE and KNN the factual-counterfactual pairs are mutually independent, so COLA is effective for instance-based CE even though our formulation in equation (1) is stated more generally to also cover group and distributional CE. Notably, $\text{CF-}p_{\text{OT}}$ sometimes outperforms $\text{CF-}p_{\text{Ect}}$, because it identifies the key features to modify through a more theoretically grounded route, whereas $\text{CF-}p_{\text{Ect}}$ leans on CE algorithm-dependent knowledge whose value for FA is not well justified in general. A gap between $\text{CF-}p_{\text{Ect}}$, $\text{CF-}p_{\text{OT}}$, and the optimum remains for KNN, and identifying the best alignment across settings is an open question we leave for future work.

7 CONCLUSIONS

This paper introduces COLA, a general framework for refining CE through joint-distribution-informed Shapley values that preserves the counterfactual effect while requiring fewer actions. Because it makes no structural assumptions about the predictor or the CE generator and treats the coupling, attribution, and value-assignment steps as interchangeable modules, the framework applies uniformly across model families and across instance-, group-, and distribution-level counterfactual paradigms, and our experiments across four domains, twelve model classes, and five generators indicate that its benefits carry over broadly. Natural directions for extending the same principles include richer data modalities and settings where a causal graph is available.

ETHICS STATEMENT

All authors have read and agree to abide by the ICLR Code of Ethics.⁴ This paper proposes a general method (*COLA* with p -SHAP) for refining CE. Our work is methodological and empirical; it does not involve human subjects or interventions, and we did not collect, annotate, or release any new personal data. Because the method is intended to apply broadly across models, generators, and application domains, the ethical considerations below are framed to cover the range of settings in which it might be used.

All experiments use standard, de-identified, publicly available tabular datasets (e.g., Adult/Census Income, German Credit, HELOC). We adhere to dataset licenses and terms and to the common research practice of using such data solely for benchmarking. We do not attempt re-identification or linkage, and we release no additional personal information. Our code will include options to exclude protected attributes and to enforce feasibility and immutability constraints on features, so that responsible use is supported across the diverse domains where CE arise.

³This knowledge — an exact alignment between factual and counterfactual — is available only from DiCE and KNN among all the CE methods considered, which is itself a reason to prefer an approach that does not require it.

⁴<https://iclr.cc/public/CodeOfEthics>

CE can surface biases in underlying models but do not by themselves guarantee fairness. Our method reduces the number of recommended feature edits; it does *not* authorize edits to sensitive or immutable attributes (e.g., sex, race, age), nor does it confer causal validity. We discourage using CE to “game” high-stakes systems (e.g., lending, hiring, healthcare) or to pressure individuals into unrealistic or unethical behavior changes. Because the framework is broadly applicable, any deployment in consequential domains should be accompanied by domain-expert-defined feasibility constraints, fairness audits (e.g., disparate impact or benefit across groups), human oversight, and compliance with applicable data-protection and AI regulations. We explicitly caution that our attribution is distributional, not causal, and must not be interpreted as a causal effect.

The principal risks — which apply wherever the method is deployed — include (i) misinterpreting attributions as causal, (ii) recommending infeasible or harmful edits, and (iii) enabling strategic manipulation. We mitigate these by (a) stating the non-causal scope and limitations in the paper, (b) supporting cost and feasibility masks and immutable features in the implementation, (c) reporting performance across subgroups where relevant, and (d) providing guidance for responsible use in the documentation.

We provide an anonymized implementation, detailed settings, and seeds to facilitate replication; we report model types, hyperparameters, and compute budgets, and we avoid result cherry-picking by using fixed evaluation protocols. To our knowledge, there are no conflicts of interest or sensitive sponsorships related to this submission; any funding disclosures will be provided after the double-blind review.

REPRODUCIBILITY STATEMENT

We take reproducibility seriously and provide pointers to all necessary components. The COLA framework and p-SHAP attribution are specified with pseudocode and notation in the main paper (Algorithm 1; Method and Preliminaries sections), with all theoretical assumptions and complete proofs deferred to the appendix (theorems and lemmas with line-by-line proofs).

Implementation details—hyperparameters, optimization settings, environment specifications, and random seeds—are documented in Appendix I, which also describes the full range of datasets used and the evaluation protocol (including data splits and preprocessing). The baselines (models and CE generators) and coupling choices spanning the various settings are enumerated in the experimental sections and ablations, with their configurations mirrored in the supplemental materials. The experiment code and software will be released upon acceptance; we also report hardware and runtime in Appendix I to contextualize compute requirements, and provide ablation and sensitivity studies together with the small-scale MILP benchmark setup in the appendix to support independent verification across configurations.

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A COMPARISON TO EXISTING APPROACHES

The authors of (Albini et al., 2022) proposed CF-SHAP, which uses counterfactual data points as the background distribution for Shapley. It assumes, however, that the counterfactual distribution is defined conditionally on each single data instance, which presupposes a known probabilistic alignment between every factual and every counterfactual instance. Similar assumptions appear in (Merrick & Taly, 2020; Kommiya Mothilal et al., 2021). This assumption breaks down in the many scenarios where global explanations are expected, and we note that the counterfactual-distribution setup of (Albini et al., 2022) is a special case of ours — our formulation recovers it while extending naturally to settings it cannot reach. The authors of (Kwon & Zou, 2022) proposed WeightedSHAP, which weights features rather than treating them as contributing equally; by analogy, our method weights the contributions of data points and extends straightforwardly to weighting both rows and columns. The framework of (Kommiya Mothilal et al., 2021) combines FA and CE for explainability, but its CE-based FA inherits the same assumption as (Albini et al., 2022), making it hard to generalize to group (Ley et al., 2023; 2022) or distributional CE (You et al., 2025). Another relevant work, (Sharma et al.), models counterfactuals through a system’s causal structure and time-series predictors to fairly decompose a single observed change in a system-level metric into contributions from individual inputs. Crucially, none of the above addresses the minimal-actions CE problem that is the focus of our paper, and none aims for the model- and generator-agnostic breadth we target.

The problems formulated in (Kanamori et al., 2022; Karimi et al., 2021) are close to the one investigated here, but each rests on narrower assumptions. In (Karimi et al., 2021) the cost of performing actions is minimized under a known structural causal model (SCM), which is rarely available in practice. The authors of (Kanamori et al., 2022) observe that it remains open whether existing CE methods can solve that problem, and propose an MILP-solver-based approach for linear classifiers, tree ensembles, and deep ReLU networks, building on (Ustun et al., 2019; Kanamori et al., 2020; Parmentier & Vidal, 2021). Solving an MILP is costly, however, which limits scalability; our framework instead trades exactness for a solution that scales and applies across arbitrary model families.

B $\mathcal{NP}$ -HARDNESS OF THE PROBLEM IN EQUATION (1)

Theorem B.1 below states that, unless $\mathcal{P} = \mathcal{NP}$, no scalable exact algorithm can be expected for solving the problem in general; the hardness stems not only from the possible non-linearity of f but also, more fundamentally, from the combinatorial nature of the problem. This is what motivates the broadly applicable approximate framework we develop rather than a solver tied to any particular model class.

Theorem B.1. *Problem equation (1) is generally $\mathcal{NP}$ -hard for non-trivial divergences. More specifically, it is hard even when $d = 1$ for linear models.*

Proof. Consider the *Sparse Regression (SR) with a Cardinality Constraint* problem, defined as follows. Given a matrix $\mathbf{W} \in \mathbb{R}^{m \times n}$, a target vector $\mathbf{y}^* \in \mathbb{R}^m$, and a sparsity level $K \in \mathbb{N}$, the goal is to find a vector $\mathbf{z} \in \mathbb{R}^n$ that minimizes the residual error while having at most K non-zero elements:

$$\begin{aligned} \min_{\mathbf{z}} \quad & \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \\ \text{s.t.} \quad & \|\mathbf{z}\|_0 \leq K, \end{aligned}$$

where $\|\mathbf{z}\|_0$ denotes the number of non-zero elements in $\mathbf{z}$. This problem is known to be $\mathcal{NP}$ -hard due to the combinatorial nature of selecting the subset of variables to include in the model.

We will map this SR problem to our problem equation (1) with the following settings. Let the number of features be $d = 1$ and the number of instances be n (the same as the dimension of the SR problem). Let the factual data be $\mathbf{x} = \mathbf{0} \in \mathbb{R}^n$ (the zero vector), and the target model output be $\mathbf{y}^* \in \mathbb{R}^m$ (as given in the SR problem). We define the model f as a linear function $f(\mathbf{z}) = \mathbf{W}\mathbf{z}$.

For the model output, we define $D(f(\mathbf{z}), \mathbf{y}^*) = \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2$, i.e. the Euclidean distance squared. For the instances, we define $D(\mathbf{z}, \mathbf{x}) = \|\mathbf{z} - \mathbf{x}\|_2^2 = \|\mathbf{z}\|_2^2$, since $\mathbf{x} = \mathbf{0}$. We set the maximum allowed feature changes C to be equal to k (the sparsity level from the SR problem). The large constant M can be any sufficiently large positive number, for example, $M \geq \max_i |z_i|$.

Given that $\mathbf{x} = \mathbf{0}$, constraints equation (1d) and equation (1e) are simplified to:

$$\begin{aligned} z_i &\leq 0 \cdot (1 - c_i) + M c_i = M c_i, \\ z_i &\geq 0 \cdot (1 - c_i) - M c_i = -M c_i, \quad \forall k = 1, \dots, n. \end{aligned}$$

This means that if $c_i = 0$, then $z_i \leq 0$ and $z_i \geq 0$, so $z_i = 0$. If $c_i = 1$, then $z_i \in [-M, M]$. The constraint equation (1c) becomes $\sum_{k=1}^n c_i \leq k$.

Our problem equation (1) thus becomes:

$$\min_{\mathbf{c}, \mathbf{z}} \quad \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \tag{11a}$$

$$\text{s.t.} \quad \|\mathbf{z}\|_2^2 \leq \epsilon, \tag{11b}$$

$$\sum_{k=1}^d c_i \leq k, \tag{11c}$$

$$z_i = 0, \quad \text{if } c_i = 0, \tag{11d}$$

$$z_i \in [-M, M], \quad \text{if } c_i = 1, \tag{11e}$$

$$c_i \in \{0, 1\}, \quad \forall k = 1, \dots, n. \tag{11f}$$

In this formulation, the variables c_i indicate whether the variable z_i is allowed to change ($c_i = 1$) or not ($c_i = 0$). The constraints enforce that $z_i = 0$ when $c_i = 0$, mirroring the sparsity constraint in the SR problem. The constraint $\sum_{k=1}^n c_i \leq k$ ensures that at most k features can change, matching the sparsity level. The objective function is identical to that of SR.

Since our problem formulation directly mirrors the SR problem with a cardinality constraint, which is known to be $\mathcal{NP}$ -hard, solving Problem equation (1) is at least as hard as solving the SR problem. Therefore, Problem equation (1) is $\mathcal{NP}$ -hard even when $d = 1$, the model f is linear, and the divergence functions D are standard Euclidean distances. $\square$

C PROOF OF THEOREM 4.1: p -SHAP TOWARDS COUNTERFACTUAL EFFECT

Let $\mathbf{x} = \{\mathbf{x}_i\}_{i=1}^n \in \mathbb{R}^{n \times d}$ be the set of factual data points with associated probability weights $\mu_i \geq 0$ such that $\sum_{i=1}^n \mu_i = 1$, and let $\mathbf{r} = \{\mathbf{r}_j\}_{j=1}^m \in \mathbb{R}^{m \times d}$ be the set of counterfactual data points with associated probability weights $\nu_j \geq 0$ such that $\sum_{j=1}^m \nu_j = 1$. Let $\mathbf{p}_{\text{OT}} \in \mathbb{R}^{n \times m}$ be the OT plan between $\mathbf{x}$ and $\mathbf{r}$ that minimizes the expected transportation cost:

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\mu, \nu)} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2,$$

where $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ is the set of joint distributions satisfying the marginal constraints $\sum_{j=1}^m p_{ij} = \mu_i$ for all i and $\sum_{i=1}^n p_{ij} = \nu_j$ for all j . The theorem below establishes that feature attributions align with the expected costs of feature modifications, yielding action plans that are cost-efficient in achieving counterfactual outcomes — a guarantee that holds generally under only the stated Lipschitz assumption, without reference to any specific model family or CE generator.

Theorem C.1 (Theorem 4.1 in the main text). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\boldsymbol{\pi} \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{OT}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \quad \forall \mathbf{p} \in \Pi.$$

Namely, $\mathbf{p}_{OT}$ minimizes the upper bound of $W_1(f(\mathbf{x}), \mathbf{y}^*)$, where the upper bound is based on the expected feature modification cost.

Proof. Since the model f is Lipschitz continuous with constant L , for any $\mathbf{x}_i \in \mathbb{R}^d$ and $\mathbf{r}_j \in \mathbb{R}^d$, it holds that:

$$|f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Multiplying both sides of the inequality by $p_{ij}^{OT} \geq 0$, we obtain:

$$p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Summing both sides over all $i = 1, \dots, n$ and $j = 1, \dots, m$, we have:

$$\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Let us denote $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$ and $E_d = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2$. The inequality then becomes:

$$E_f \leq L E_d.$$

To further bound E_d , we apply the Cauchy-Schwarz inequality. Observe that the weights p_{ij}^{OT} are non-negative and satisfy $\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} = 1$ because $\mathbf{p}_{OT}$ is a probability distribution over the joint space of $\mathbf{x}$ and $\mathbf{r}$. The Cauchy-Schwarz inequality states that for any real-valued functions a_{ij} and b_{ij} ,

$$\left(\sum_{i,j} a_{ij} b_{ij} \right)^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right).$$

Setting $a_{ij} = \sqrt{p_{ij}^{OT}}$ and $b_{ij} = \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2$, we have:

$$E_d = \sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} \sqrt{p_{ij}^{OT}} \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} a_{ij} b_{ij}.$$

Applying the Cauchy-Schwarz inequality:

$$E_d^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right) = \left(\sum_{i,j} p_{ij}^{OT} \right) \left(\sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 \right).$$

Since $\sum_{i,j} p_{ij}^{OT} = 1$, this simplifies to:

$$E_d^2 \leq \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2.$$

Taking the square root of both sides, we obtain:

$$E_d \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Substituting back into the inequality for E_f , we have:

$$E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Note that the 1-Wasserstein divergence is no more than E_f^5 , then,

$$W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*| \leq E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Therefore, the 1-Wasserstein divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$ is bounded by the Lipschitz constant L times the square root of the expected transportation cost under $\mathbf{p}_{\text{OT}}$. Since $\mathbf{p}_{\text{OT}}$ minimizes the expected transportation cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$ over all feasible transport plans in $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$, we have

$$\sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

Hence the conclusion follows. $\square$

D PROOF OF THEOREM 4.2: INTERVENTIONAL EFFECT

Theorem D.1 (Theorem 4.2 in the main text). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Proof. Let $p(\mathbf{x}, \mathbf{r})$ be the joint probability of $\mathbf{x}$ and $\mathbf{r}$ obtained from A_{Prob} . Under the intervention $do(\mathbf{r}_S = \mathbf{x}_{i,S})$, the features in S are set to $\mathbf{x}_{i,S}$, and the features in $\mathcal{F} \setminus S$ remain distributed according to their marginal distribution $p(\mathbf{r}_{\mathcal{F} \setminus S})$. Therefore, the expected value of $f(\mathbf{r})$ under the intervention is:

$$\mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S}.$$

Remark that by the definition of $v^{(i)}(S)$,

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r} | \mathbf{x}_i)}[f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}[f(\mathbf{r})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S} - \mathbb{E}[f(\mathbf{r})],$$

such that

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Hence the conclusion. $\square$

The value function $v^{(i)}(S)$ captures the expected value of $f(\mathbf{r})$ when we intervene by setting the features in S to $\mathbf{x}_{i,S}$, denoted $do(\mathbf{r}_S = \mathbf{x}_{i,S})$. This intervention holds independently of any predefined joint probability distribution $p(\mathbf{x}, \mathbf{r})$, which is what lets the interpretation apply across couplings. Consequently, the expression $\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S)$ represents the combined effect of the base expected value of $f(\mathbf{r})$ and the additional causal impact of the attribution $v^{(i)}(S)$.

⁵Note that $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$. Because both π and $\mathbf{p}_{\text{OT}}$ denote joint probability of $\mathbf{x}$ and $\mathbf{r}$, however, π makes the summation the minimum for the W_1 term across all possible joint distributions, whereas $\mathbf{p}_{\text{OT}}$ does not.

E PROOF OF THEOREM 5.1: COUNTERFACTUAL PROXIMITY

Theorem E.1 (Theorem 5.1 in the main text). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is strictly closer (or equal) to the factual input $\mathbf{x}$ compared to the aligned counterfactual proposal $\mathbf{q}$, ensuring that the refinement process introduces no additional divergence.

Proof. Let $\mathbf{q}$ be the aligned reference matrix generated by $A_{\text{value}}(\mathbf{r}, \mathbf{p})$. For the elements where $c_{ik} = 1$ (modified elements), $z_{ik} = q_{ik}$. For elements where $c_{ik} = 0$, $z_{ik} = x_{ik}$. Therefore, we can write:

$$(z_{ik} - x_{ik})^2 = \begin{cases} (q_{ik} - x_{ik})^2, & \text{if } c_{ik} = 1; \\ 0, & \text{if } c_{ik} = 0. \end{cases} \quad (12)$$

The squared Frobenius norms with respect to $\mathbf{q}$ and $\mathbf{z}$ are computed as follows:

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2, \\ \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2. \end{aligned}$$

And,

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2 \\ &\stackrel{(i)}{=} \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{(i,k): c_{ik}=1} (q_{ik} - x_{ik})^2 \\ &= \sum_{(i,k): c_{ik}=0} (q_{ik} - x_{ik})^2 \geq 0, \end{aligned}$$

where the equality (i) holds because of equation (12).

Since the difference $\|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 \geq 0$, it follows that:

$$\|\mathbf{z} - \mathbf{x}\|_F^2 \leq \|\mathbf{q} - \mathbf{x}\|_F^2.$$

Taking square roots, we get:

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

Hence the conclusion. $\square$

Corollary E.2 (Corollary 5.2 in the main text). *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

recovering the intuition that the refined counterfactual is at least as close to the factuals as the original counterfactual set reordered by σ .

Proof. Under the assumptions $n = m$ and $\varepsilon = 0$, the optimal transport plan $\mathbf{p}$ becomes a permutation matrix associated with a bijection σ . Consequently, the value assignment algorithm A_{value} (whether maximizing or averaging) assigns $\mathbf{q}_i = \mathbf{r}_{\sigma(i)}$ for every instance i . Substituting $\mathbf{q}$ with the permuted matrix $\mathbf{r}_\sigma$ into the general bound established in Theorem 5.1 (i.e., $\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F$) directly yields the result. $\square$

F ANALYSIS OF COMPUTATIONAL COMPLEXITY OF COLA

We analyze the computational complexity of COLA as follows, treating each module in the generality with which it is used.

First, consider A_{Prob} . If the alignment between $\mathbf{x}$ and $\mathbf{r}$ is known a priori, then A_{Prob} merely constructs the matrix $\mathbf{p}$ from that prior knowledge, which takes $\mathcal{O}(n \times m)$. Otherwise — the more common situation — we solve OT to obtain $\mathbf{p}$, which by the Sinkhorn–Knopp algorithm takes $\mathcal{O}(n \times m \times \log(1/\varepsilon))$.

Then we analyze A_{Shap} . For each subset, the model is evaluated on all n data points, leading to $\mathcal{O}(n)$ evaluations per subset. Incorporating baseline values from the reference data $\mathbf{r}$ involves replacing the values of certain features with their corresponding baseline values. This operation is $\mathcal{O}(m)$ because it requires accessing the baseline values from the reference table $\mathbf{r}$ for each of the d features. If we assume that the reference values can be precomputed and accessed in constant time, then the complexity of incorporating these values can be considered as $\mathcal{O}(d)$. The number of M_{Shap} subsets results in M_{Shap} model evaluations. Combining the above steps, the complexity of A_{Shap} is $\mathcal{O}(n \times d \times M_{\text{Shap}})$.

The normalization in line 4 of COLA takes $\mathcal{O}(n \times d)$.

To compare the complexities of the two algorithms, $A_{\text{Value}}^{\max}$ and $A_{\text{Value}}^{\text{avg}}$, we analyze each algorithm step-by-step. For $A_{\text{Value}}^{\max}$, for each row $\mathbf{x}_i$ in the data table, we need to (1) compute the probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, which involves $\mathcal{O}(m)$ operations per row, (2) identify the row $\mathbf{r}_j$ in the reference data with the highest probability p_{ij} , which involves $\mathcal{O}(m)$ operations per row, and (3) assign $q_{ik} = r_{\tau(i),k}$ where $\tau(i) = \arg \max_j p_{ij}$, which involves $\mathcal{O}(d)$ operations per row. Since there are n rows in the data table, the total complexity for $A_{\text{Value}}^{\max}$ is $\mathcal{O}(n \times (m + m + d)) = \mathcal{O}(n \times (2m + d)) = \mathcal{O}(n \times m + n \times d) = \mathcal{O}(n \times (m + d))$.

For $A_{\text{Value}}^{\text{avg}}$, for each row $\mathbf{x}_i$ in the data table, we need to 1) compute the probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, which involves $\mathcal{O}(m)$ operations per row, 2) compute the sum $\sum_{j'=1}^m p_{ij'}$, which involves $\mathcal{O}(m)$ operations per row, and 3) calculate the weighted average $q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}$, which involves $\mathcal{O}(m \times d)$ operations per row. Since there are n rows in the data table, the total complexity for $A_{\text{Value}}^{\text{avg}}$ is $\mathcal{O}(n \times (m + m + m \times d)) = \mathcal{O}(n \times (2m + m \times d)) = \mathcal{O}(n \times (m + m \times d)) = \mathcal{O}(n \times m \times (1 + d)) = \mathcal{O}(n \times m \times d)$.

For lines 6–16, the entire complexity is straightforwardly $\mathcal{O}(n \times d) + \mathcal{O}(C) = \mathcal{O}(n \times d)$ due to the fact $C \leq n \times d$.

Therefore, the complexity of COLA using $A_{\text{Value}}^{\max}$ equals

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(n(m + d)) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nm) + \mathcal{O}(nd), \end{aligned}$$

and the complexity of COLA using $A_{\text{Value}}^{\text{avg}}$ equals

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(nmd) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nmd). \end{aligned}$$

Hence the complexity of COLA with respect to n , m , d , and the regularization parameter ε of entropic OT is

$$\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N,$$

where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ if $A_{\text{Value}}^{\max}$ is used and $N = \mathcal{O}(nmd)$ if $A_{\text{Value}}^{\text{avg}}$ is used.

G AN MILP FORMULATION OF EQUATION (1) WITH MEAND

In this section we derive a global optimality benchmark for the case of a known alignment between factual and counterfactual when solving equation (1) with D being MeanD. This serves as an exact reference point against which the broadly applicable COLA framework can be calibrated, namely

$$D(f(\mathbf{z}), \mathbf{y}^*) = \left| \frac{1}{n} \sum_{i=1}^n f(\mathbf{z}_i) - \bar{y}^* \right|,$$

with $\bar{y}^* = \frac{1}{m} \sum_{j=1}^m y_j^*$. Since COLA is used, we have $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$, and $\mathbf{z}$ stays closer to $\mathbf{x}$ than $\mathbf{r}$, hence equation (1b) is dropped. The formulation of equation (1) then becomes:

$$\min_{\mathbf{c}, \mathbf{z}} \left| \sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* \right|, \quad (13a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (13b)$$

$$z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik}), \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (13c)$$

Note that the original constraints equation (1d) and equation (1e) merge to be equation (13c), because CF- p_{Ect} is imposed to be used. That is, for any $\mathbf{x}_i$, there is an exact $\mathbf{r}_j$ serves as its reference in A_{Shap} and A_{Value} , such that x_{ik} ($k = 1, 2, \dots, d$) either stays unchanged or can be changed to r_{jk} . Therefore $z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik})$ of which the value depends on the binary variable c_{ik} .

Due to the known alignment between any $\mathbf{x}_i$ and its corresponding $\mathbf{r}_j$, $\mathbf{q}$ is determined (also, both $A_{\text{Values}}^{\max}$ and $A_{\text{Values}}^{\text{avg}}$ return the same $\mathbf{q}$). For any data point i and any feature set $S \subseteq \mathcal{F}$, let $\mathbf{z}_{iS}$ denotes the solution $\mathbf{z}_i$ where we have all features $k \in S$ changed to q_{ik} , and the other features $h \in \mathcal{F} \setminus S$ stays x_{ih} . Hence the set of $\mathbf{z}_{iS}$ ($S \subseteq \mathcal{F}$) composes the domain of all possible values of $\mathbf{z}$. Define a corresponding scaler variable for any $\mathbf{z}_{iS}$:

$$g_{iS} = f(\mathbf{z}_{iS}) - \bar{y}^*.$$

Then, for any $\mathbf{z}_i$ in equation (13), the value of the term $\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^*$ can be represented by a binary variable a_{iS} together with the scaler g_{iS} , namely,

$$\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* = \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS}.$$

The optimization problem equation (13) is hence reformulated as a mixed integer programming below.

$$\min_{\mathbf{a}, \eta} \quad \eta, \quad (14a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \leq \eta, \quad (14b)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \geq -\eta, \quad (14c)$$

$$\sum_{S \subseteq \mathcal{F}} a_{iS} = 1, \quad i = 1, \dots, n, \quad (14d)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} |S| a_{iS} \leq C. \quad (14e)$$

Minimizing η under the two constraints equation (14b) and equation (14c) is equivalent to minimizing the objective function of equation (13). The constraints in equation (14d) guarantees that each data point i is subject to one and only one feature modification plan a_{iS} for a specific S ($S \subseteq \mathcal{F}$). The constraint equation (14e) corresponds to equation (13b). Solving equation (14) yields the theoretical optimum for COLA under a known factual-counterfactual alignment, which serves as the exact reference used in Figure 4 of Section 6.

H EXTENDED NUMERICAL RESULTS

Across Figures 5–8, CF- p_{Uni} generally outperforms RB- p_{Uni} . Second, considering RB- p_{OT} and CF- p_{OT} , which also differ only in A_{Shap} , the latter consistently outperforms the former. We therefore conclude, broadly across these settings, that *RB-SHAP is not suitable for FA in CE*.

We analyze how different Shapley methods affect FA, corresponding to lines 3–4 in COLA. The Shapley methods fall into two categories. First, consider RB- p_{Uni} and CF- p_{Uni} , which differ only

in A_{Shap} ; across Figures 5–8, $\text{CF-}p_{\text{Uni}}$ generally performs better than $\text{RB-}p_{\text{Uni}}$. Second, consider $\text{RB-}p_{\text{OT}}$ and $\text{CF-}p_{\text{OT}}$, again differing only in A_{Shap} ; the latter consistently outperforms the former. This pattern holds uniformly across the datasets, models, and generators examined, reinforcing the same conclusion.

Besides FA, the other equally important step of COLA is line 5, which uses the joint probability $p(\mathbf{x}, \mathbf{r})$ to compose the matrix $\mathbf{q}$, indicating the direction in which the factual x should change its features to move toward the target model outcome. We observe in Figures 5–8 that $\text{CF-}p_{\text{OT}}$ consistently outperforms all other methods throughout every experiment. All three of $\text{CF-}p_{\text{Uni}}$, $\text{CF-}p_{\text{Rnd}}$, and $\text{CF-}p_{\text{OT}}$ supply a joint probability $\mathbf{p}$ in the common case where the exact factual–counterfactual alignment is unknown, yet their performance differs substantially. Merely knowing the CE (and its marginal distribution) is insufficient in general.

OT proves especially useful whenever alignment information between factual and counterfactual instances is missing or inaccurate — the prevailing situation for group and distributional CE. Even when a CE algorithm explicitly matches each factual instance to a corresponding counterfactual, it is hard to justify that this known alignment is performance-optimal, as Figure 4 in Section 6 supports.

Note that p_{OT} need not be the true joint distribution of $\mathbf{x}$ and $\mathbf{r}$ in a data-generation sense. Rather, its role is to guide COLA to treat $\mathbf{x}$ and $\mathbf{r}$ jointly for both FA and CE. Furthermore, the QDA column in Figure 5 shows the stability of OT-based methods, whereas the others diverge markedly from the target. We emphasize, however, that COLA is *not restricted to using OT* as A_{Prob} ; OT is a strong default within a general slot. As Figure 4 indicates, even the best known $\mathbf{p}$ leaves a non-negligible gap to global optimality, so devising a better A_{Prob} is a promising direction that the generality of the framework directly opens up.

I EXPERIMENTS REPRODUCIBILITY

The experiments are conducted on a high-performance computing (HPC) cluster running four nodes in parallel (one per dataset), each node equipped with two Intel Xeon Processor 2660v3 (10 core, 2.60GHz) and 128 GB memory. Each node runs for roughly 5–10 hours depending on the size of the dataset. The experiments can also be reproduced on a laptop, though generally at a higher time cost than on an HPC cluster; the modest resource requirements make the study broadly reproducible.

Across all four datasets, numerical features are standardized and categorical features are handled by either label encoding or one-hot encoding. In practice we observed no appreciable difference between the two encoding schemes in terms of COLA’s performance, suggesting the framework is insensitive to this preprocessing choice. The train–test split is 7 : 3 throughout.

The optimality baseline shown in Figure 4 is solved with Gurobi 11.0.2 ([gur](#)). Reproducing it requires a Gurobi license; alternatively, open-source operations-research libraries such as Google OR-Tools ([goo](#)) can be used. As noted, solving the MILP in Appendix G is computationally expensive and hence practical only for small-scale datasets such as German Credit; for larger datasets the number of features used would need to be reduced. This computational limit applies only to the exact baseline — COLA itself scales to all datasets considered.

The model hyperparameters are as follows. Bagging, GP, RBF, RndForest, AdaBoost, GradBoost, and QDA are scikit-learn models ([skl](#)) with all hyperparameters kept at their defaults. DNN, SVM, RBF, and LR are implemented in PyTorch ([Paszke et al., 2019](#)); the DNN has three layers and the SVM uses a linear kernel. XGBoost ([Chen & Guestrin, 2016](#)) and LightGBM ([Ke et al., 2017](#)) are used through their scikit-learn interfaces, again with default hyperparameters. Keeping default settings throughout avoids per-model tuning and indicates that the reported gains are not the product of configuration-specific optimization.

THE USE OF LARGE LANGUAGE MODELS (LLMs)

We used large language models (LLMs) only as general-purpose assistance tools. Their role was confined to grammar, wording, and light copy-editing of author-written text. LLMs did not contribute to research ideation, modeling choices, experimental design, or results. All algorithms, proofs, datasets, and analyses were created and verified by the authors. Any LLM-suggested phrasing was

1188 reviewed and edited before inclusion in the paper. The authors take full responsibility for all content;
1189 LLMs are neither authors nor contributors. This disclosure complies with the ICLR policy on LLM
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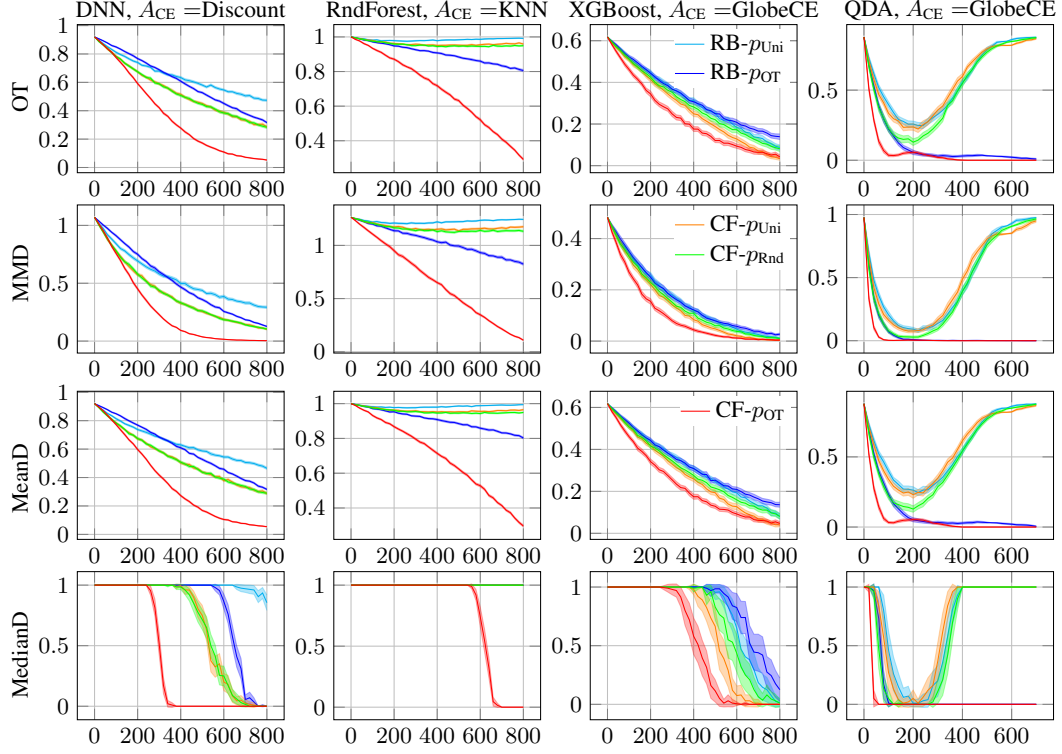


Figure 5: *[HELOC]* $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. allowed actions C , across the full grid of models and generators for this dataset. Experiments use 100 runs, with shaded regions giving the 99.9% confidence intervals. The legends apply to all plots. $A_{\text{Value}}^{\text{avg}}$ is used.

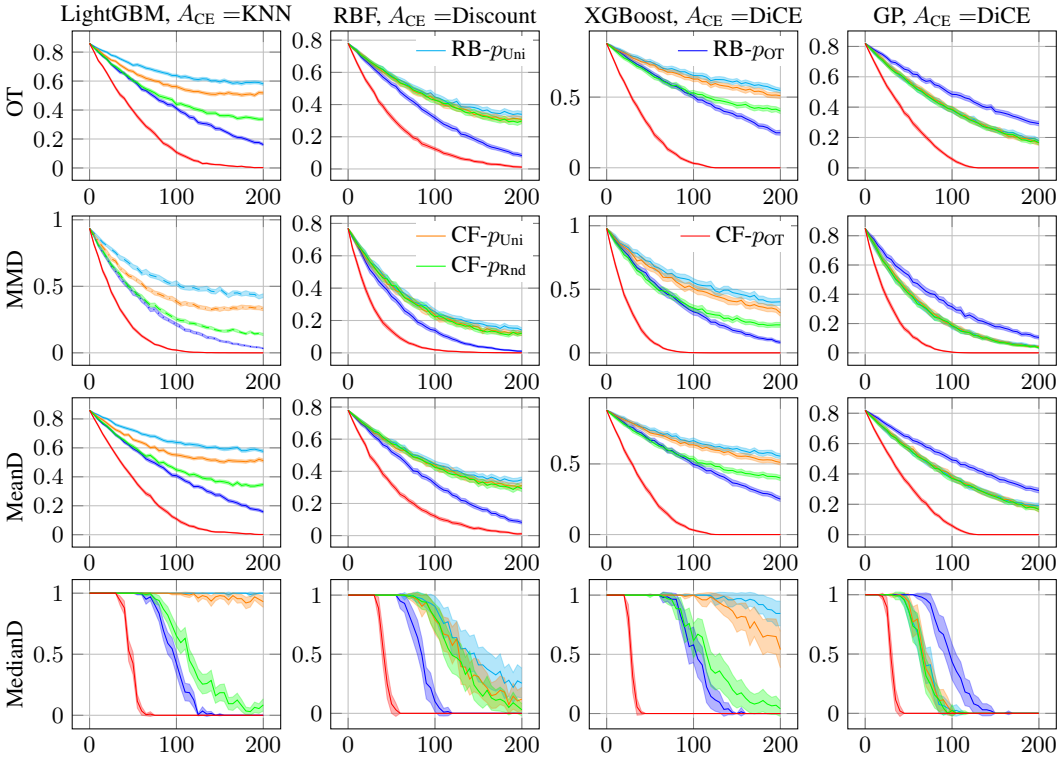


Figure 6: *[German Credit]* $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. allowed actions C , across the full grid of models and generators for this dataset. Experiments use 100 runs, with shaded regions giving the 99.9% confidence intervals. The legends apply to all plots. $A_{\text{Value}}^{\text{avg}}$ is used.

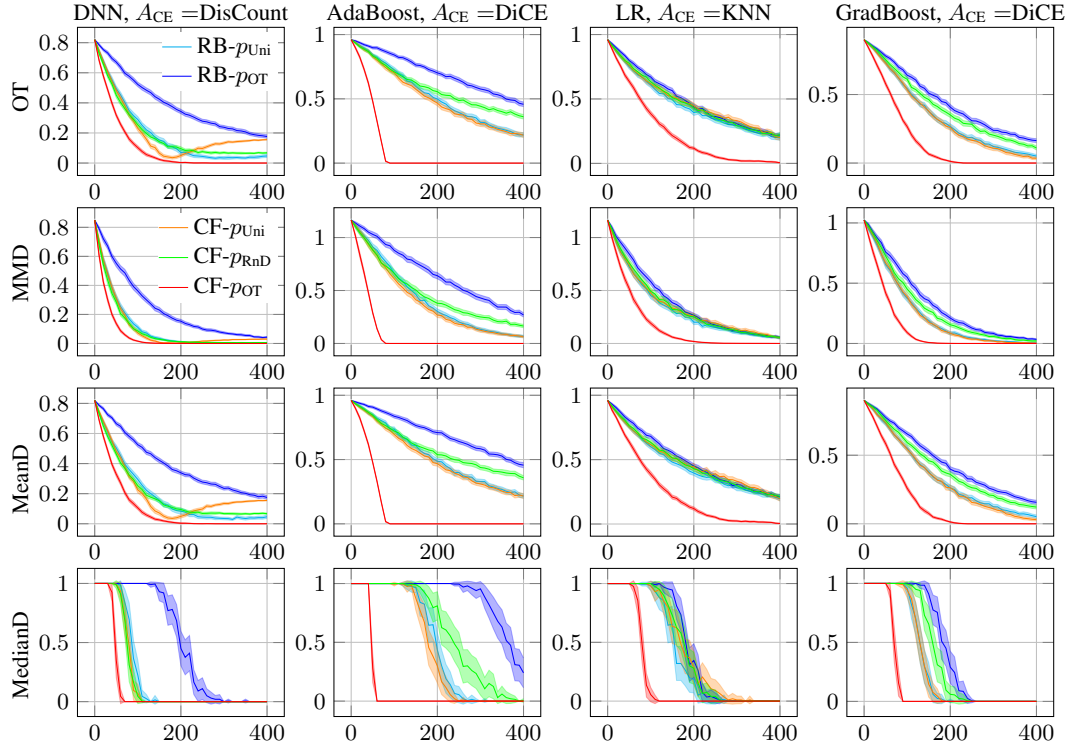


Figure 7: [Hotel Bookings] $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. allowed actions C , across the full grid of models and generators for this dataset. Experiments use 100 runs, with shaded regions giving the 99.9% confidence intervals. The legends apply to all plots. $A_{\text{Value}}^{\max}$ is used.

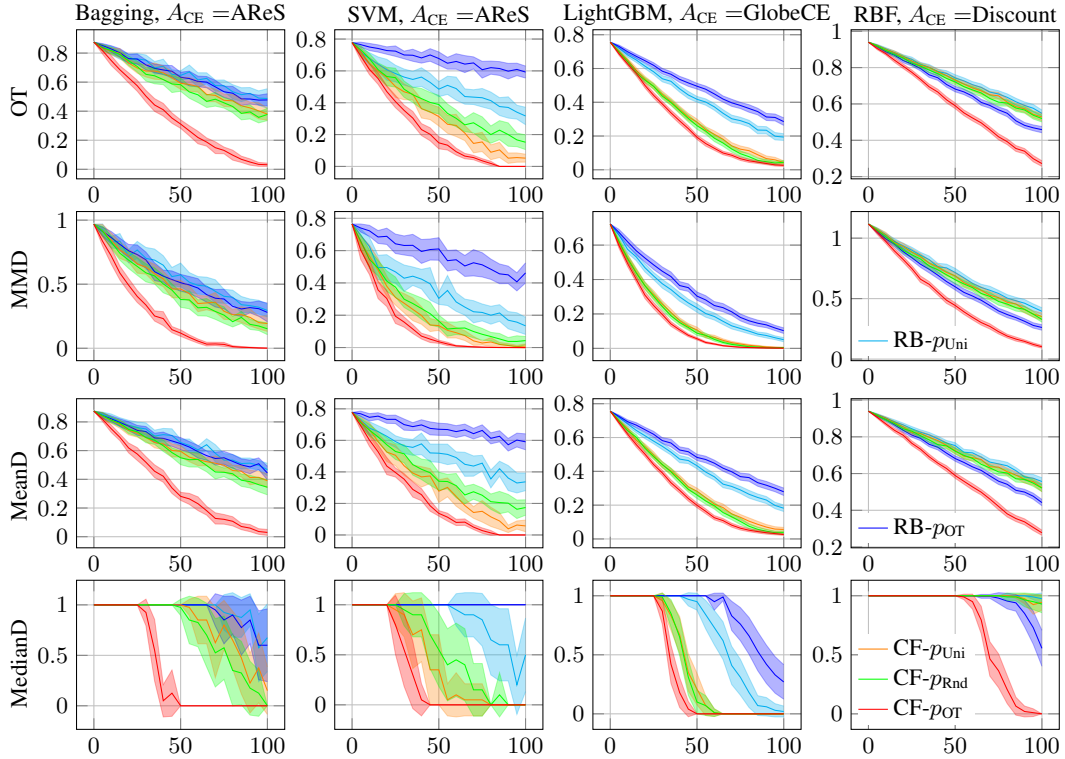


Figure 8: [COMPAS] $D(f(\mathbf{z}), \mathbf{y}^*)$ vs. allowed actions C , across the full grid of models and generators for this dataset. Experiments use 100 runs, with shaded regions giving the 99.9% confidence intervals. The legends apply to all plots. $A_{\text{Value}}^{\max}$ is used.